IB Biology SL — 2025 Syllabus

THEME C - INTERACTION & INTERDEPENDENCE

Exam: Paper 1A (MCQ) · Paper 1B (Data) · Paper 2A (Short answer) · Paper 2B (Extended response)

WHAT'S IN THIS GUIDE

C1.1 Enzymes & Metabolism — C1.2 Cell Respiration — C1.3 Photosynthesis — C2.2 Neural Signalling — C3.1 Integration of Body Systems — C3.2 Defence Against Disease — C4.1 Populations & Communities — C4.2 Transfers of Energy & Matter

This theme asks: how do biological systems interact — within cells, within organisms, and between organisms? It covers the molecular machinery of metabolism, the nervous and endocrine systems, immunity, and ecology.

High-frequency exam topics: enzyme inhibition, ATP cycle, aerobic respiration stages, Calvin cycle, action potential, reflex arc, immune response, population growth curves, energy transfer efficiency.

C1.1 — Enzymes & Metabolism

Induced fit · Activation energy · Factors affecting rate · Inhibition

Enzymes are biological catalysts — proteins that speed up chemical reactions by lowering activation energy without being consumed. They are essential for every metabolic process in the cell.

KEY DEFINITIONS

Metabolism: the sum of ALL chemical reactions occurring in a living organism. Includes both anabolic and catabolic reactions.

Anabolism: metabolic reactions that BUILD UP complex molecules from simpler ones. Require energy input. Example: protein synthesis (amino acids → polypeptide), photosynthesis (CO_2 + H_2O → glucose).

Catabolism: metabolic reactions that BREAK DOWN complex molecules into simpler ones. Release energy. Example: cellular respiration (glucose → CO_2 + H_2O + ATP), digestion.

Active site: the specific region of an enzyme where the substrate binds and the reaction occurs. Its shape is complementary to the substrate (not identical — the fit is induced).

Induced fit model: when the substrate binds, the enzyme changes its shape to better accommodate the substrate — the active site 'moulds around' the substrate. More accurate than the lock-and-key model (which implied a rigid, pre-formed fit).

Activation energy: the minimum energy needed for a reaction to occur — the 'energy hill' that must be climbed. Enzymes lower this barrier — more substrate molecules have sufficient energy to react → faster reaction rate.

Cofactor: a non-protein molecule required for enzyme activity. Inorganic cofactors (e.g. metal ions: Fe^{2+} in haemoglobin, Zn^{2+} in carbonic anhydrase). Organic cofactors = coenzymes (e.g. NAD^+, FAD — electron carriers in respiration).

Factor	Effect on enzyme activity	Explanation / key details
Temperature	Increases rate up to optimum (T_{opt}) → sharp decrease above optimum	Below T_{opt}: lower kinetic energy, fewer collisions. At T_{opt}: maximum rate. Above T_{opt}: thermal energy breaks H-bonds and hydrophobic interactions holding 3D shape → DENATURATION → shape of active site changes → substrates no longer bind. Most human enzymes: T_{opt} ≈ 37°C.
pH	Optimum pH for each enzyme — activity falls at extremes	Each enzyme has a specific optimal pH (e.g. pepsin: pH 2; amylase: pH 7; arginase: pH 9). Extreme pH denatures enzyme by disrupting ionic bonds and H-bonds → changes active site shape.
Substrate concentration	Increases rate until all enzyme active sites are saturated — then plateaus (Vmax)	Initially: more substrate → more enzyme-substrate collisions → faster rate. At saturation: all active sites occupied → rate cannot increase further regardless of more substrate. Only adding more enzyme can increase Vmax.
Enzyme concentration	Increases rate proportionally (if substrate is in excess)	More enzyme molecules = more active sites available for substrate to bind.
Competitive inhibitor	Reduces rate (reversible — overcome by excess substrate)	Inhibitor molecule has similar shape to substrate → binds to ACTIVE SITE → blocks substrate from binding. Compete for the same site. Increase substrate concentration → out-compete the inhibitor. Example: methanol poisoning treated with ethanol (competes with methanol for alcohol dehydrogenase).
Non-competitive inhibitor	Reduces rate (not overcome by excess substrate)	Inhibitor binds to ALLOSTERIC SITE (different from active site) → causes CONFORMATIONAL CHANGE of enzyme → active site shape altered → substrate cannot bind properly → reduced activity. Adding more substrate does NOT reverse this.

WATCH OUT / COMMON MISTAKE

Competitive vs non-competitive inhibition: KEY DISTINCTION — competitive inhibitor binds to ACTIVE SITE and can be overcome by excess substrate. Non-competitive binds to ALLOSTERIC SITE and CANNOT be overcome by excess substrate (Vmax is reduced).

Enzymes are NOT consumed — they are released after the reaction and can catalyse more reactions. They change the RATE, not the equilibrium position of the reaction.

Denaturation at high temperature: the enzyme is not 'killed' — it is unfolded. The active site shape changes so substrates cannot bind. The primary structure (peptide bonds) is NOT broken.

C1.2 — Cell Respiration

ATP · Glycolysis · Anaerobic · Krebs cycle · Electron transport chain

Cell respiration is the process of releasing chemical energy from organic molecules (primarily glucose) and transferring it to ATP — the cell's universal energy currency. It occurs in all living cells.

KEY DEFINITIONS

ATP (adenosine triphosphate): the universal energy currency of the cell. Consists of adenosine + 3 phosphate groups. The terminal phosphate bond stores energy that is released when hydrolysed: ATP + H_2O → ADP + P_i + energy (~30 kJ/mol).

ATP cycle: ATP is constantly synthesised (from ADP + P_i using energy from respiration) and hydrolysed (releasing energy for cellular work). A cell recycles its ATP supply thousands of times per day.

NAD^+ and FAD: coenzymes that act as electron carriers in respiration. NAD^+ is reduced to NADH (accepts 2 electrons + 1 proton). FAD is reduced to $FADH_2$. They carry high-energy electrons to the electron transport chain.

Stage	Location	Inputs	Outputs	ATP yield (net)
Glycolysis	Cytoplasm (cytosol)	1 glucose (6C)	2 pyruvate (3C each), 2 NADH, 4 ATP (2 used → net 2 ATP)	Net 2 ATP
Link reaction (Pyruvate oxidation)	Mitochondrial matrix	2 pyruvate (×2 per glucose)	2 acetyl-CoA (2C), 2 CO_2, 2 NADH	0 ATP (per glucose)
Krebs cycle	Mitochondrial matrix	2 acetyl-CoA (×2 per glucose)	4 CO_2, 6 NADH, 2 $FADH_2$, 2 ATP (by substrate-level phosphorylation)	2 ATP
Oxidative phosphorylation / ETC	Inner mitochondrial membrane (cristae)	10 NADH, 2 $FADH_2$, O_2	H_2O, ~32–34 ATP (chemiosmosis)	~32–34 ATP
TOTAL (aerobic)	—	Glucose + O_2	CO_2 + H_2O + ATP	~36–38 ATP
Anaerobic (animals)	Cytoplasm only	Glucose (+ NADH)	Lactate, 2 ATP (NADH reoxidised to NAD^+)	2 ATP
Anaerobic (yeast/plants)	Cytoplasm only	Glucose (+ NADH)	Ethanol + CO_2, 2 ATP (NADH reoxidised)	2 ATP

Oxidative Phosphorylation — How the ETC Makes ATP

NADH and $FADH_2$ (from glycolysis, link reaction, and Krebs) deliver high-energy electrons to the electron transport chain (ETC) on the INNER MITOCHONDRIAL MEMBRANE.

Electrons pass along protein complexes (I, II, III, IV) — releasing energy at each step. This energy is used to PUMP PROTONS (H^+) from the matrix into the INTERMEMBRANE SPACE.

A proton gradient builds up (high [H^+] in intermembrane space, low in matrix). This is the PROTON MOTIVE FORCE (pmf).

Protons flow BACK INTO THE MATRIX through ATP synthase (Complex V) — like water turning a turbine. This drives ATP synthesis from ADP + P_i (CHEMIOSMOSIS).

At the end of the chain, electrons combine with O_2 (final electron acceptor) and H^+ to form WATER. This is why aerobic respiration REQUIRES oxygen.

Key point: NADH yields ~2.5 ATP; $FADH_2$ yields ~1.5 ATP (enters chain at a lower point). This is why NADH > $FADH_2$ in ATP value.

Anaerobic Respiration — Why Regenerate NAD^+?

Glycolysis requires NAD^+ to accept electrons. If all NAD^+ becomes NADH, glycolysis STOPS — no more ATP can be produced.

In aerobic respiration: NADH delivers electrons to the ETC → oxidised back to NAD^+ → glycolysis continues.

In anaerobic respiration: no O_2 → ETC cannot work → NADH cannot be oxidised by the ETC → NAD^+ must be regenerated by ANOTHER route:

ANIMALS/YEAST/BACTERIA: NADH + pyruvate → NAD^+ + lactate (animals) OR ethanol + CO_2 (yeast). NAD^+ regenerated → glycolysis continues → 2 ATP per glucose maintained.

This is why anaerobic respiration only yields 2 ATP — no Krebs or ETC possible. Glycolysis is the only ATP-producing stage.

EXAM TIP

ATP yield: IB SL does NOT require you to memorise exact yield of each ETC step. Know: glycolysis = 2 net ATP, Krebs = 2 ATP, ETC = ~32-34 ATP, TOTAL ≈ 36-38 ATP. Anaerobic = 2 ATP.

Location is critical: Glycolysis = CYTOPLASM. Link reaction + Krebs = MITOCHONDRIAL MATRIX. ETC = INNER MITOCHONDRIAL MEMBRANE (cristae).

Why is O_2 the final electron acceptor? It is highly electronegative — it readily accepts electrons. Without O_2, the ETC backs up — electrons cannot be passed along — NADH cannot be reoxidised.

C1.3 — Photosynthesis

Light-dependent reactions · Calvin cycle · Chromatography · Action/absorption spectra · Limiting factors

Photosynthesis converts light energy into chemical energy (glucose). It occurs in chloroplasts and has two main stages: the light-dependent reactions (in thylakoid membranes) and the light-independent reactions/Calvin cycle (in the stroma).

Stage	Location	Inputs	Key events	Outputs
Light-dependent reactions	Thylakoid membranes (grana)	Light energy, H_2O, ADP+P_i, $NADP^+$	1. Photosystem II (P680) absorbs light → excites electrons → passed to ETC. 2. Water photolysis: $2H_2O \rightarrow 4H^+ + 4e^- + O_2$ (electrons replace those lost from PSII). 3. Electrons pass along ETC → proton pumping → ATP synthesis (photophosphorylation). 4. Photosystem I (P700) absorbs light → excites electrons → reduces $NADP^+$ to NADPH.	ATP, NADPH, O_2 (released as waste)
Light-independent reactions (Calvin cycle)	Stroma of chloroplast	CO_2, ATP, NADPH (from light reactions), RuBP	1. CO_2 fixation: CO_2 + RuBP (5C) → 2× GP (3C) [catalysed by RuBisCO]. 2. Reduction: GP reduced to G3P using ATP + NADPH. 3. Regeneration: G3P → RuBP (uses ATP). 1 G3P exported per turn for organic molecule synthesis.	G3P (triose phosphate — used to make glucose, starch, amino acids, lipids), RuBP regenerated

Chromatography of Photosynthetic Pigments

Photosynthetic organisms contain MULTIPLE PIGMENTS — each absorbs different wavelengths of light, collectively broadening the range of light used for photosynthesis.

Chromatography separates pigments based on their solubility in the solvent and their affinity for the stationary phase. Less soluble pigments travel less far.

Rf value = distance moved by pigment / distance moved by solvent front. Rf is constant for a pigment with a given solvent system (useful for identification).

Pigments (in order of Rf from highest to lowest in typical IB experiment): Carotene (orange, highest Rf — most soluble), Xanthophylls (yellow), Chlorophyll a (blue-green), Chlorophyll b (green, lowest Rf — least soluble).

Chlorophyll a and b are the main photosynthetic pigments. Carotenoids are accessory pigments — absorb light wavelengths that chlorophyll cannot and transfer energy to chlorophyll.

Spectrum	Definition	Relationship
Absorption spectrum	Graph showing which wavelengths of light a pigment ABSORBS. Peaks at wavelengths most absorbed (blue ~430nm and red ~680nm for chlorophyll a; dips at green — hence green colour).	Absorption spectrum shows what the pigment can use.
Action spectrum	Graph showing the RATE OF PHOTOSYNTHESIS at different wavelengths of light. Shows which wavelengths are most effective for photosynthesis.	Should closely match the absorption spectrum — wavelengths that are absorbed drive photosynthesis most.
Relationship	The action spectrum closely matches the COMBINED absorption spectra of ALL photosynthetic pigments (chlorophylls a, b + carotenoids). Green light (550nm) is poorly absorbed — hence low action spectrum. Blue (430nm) and red (680nm) peaks match.	Confirms that light absorption drives photosynthesis. Slight differences due to multiple pigments and other cell processes.

KEY DEFINITIONS

Limiting factors: factors that prevent photosynthesis from reaching its maximum possible rate. At any given moment, the ONE factor most below its optimum is the limiting factor.

Light intensity: low light → less light energy for light reactions → less ATP and NADPH → slower Calvin cycle. Saturation point: above a certain light intensity, rate plateaus (another factor becomes limiting).

CO_2 concentration: low CO_2 → less substrate for fixation by RuBisCO → less GP → less G3P → slower Calvin cycle.

Temperature: affects Calvin cycle enzyme activity (light reactions less temperature-sensitive — driven by light, not enzymes). Low temp → slower enzyme-catalysed reactions. High temp → denaturation.

Light compensation point: the light intensity at which the rate of photosynthesis exactly equals the rate of respiration — no net gas exchange. Below this point, the plant is a net consumer of organic molecules.

EXAM TIP

Water photolysis: $2H_2O \rightarrow 4H^+ + 4e^- + O_2$. The oxygen released in photosynthesis comes from WATER — not from CO_2. This is confirmed by isotope-labelling experiments ($H_2{}^{18}O \rightarrow {}^{18}O_2$).

RuBisCO: ribulose-1,5-bisphosphate carboxylase/oxygenase — the enzyme that fixes CO_2. The most abundant enzyme on Earth. Catalyses the first step of the Calvin cycle.

Linking light-dependent and light-independent reactions: the light reactions produce ATP and NADPH used by the Calvin cycle. Without light reactions, the Calvin cycle stops (no ATP/NADPH). Without Calvin cycle, RuBP runs out and light reactions slow (no $NADP^+$ to reduce).

C2.2 — Neural Signalling

Neuron structure · Resting potential · Action potential · Synapse · EPSP

Neurons are specialised cells that transmit electrical signals over long distances. Understanding how the resting potential is maintained, how an action potential is generated, and how signals cross synapses is essential for this section.

Neuron type	Location	Function
Sensory neuron	PNS — from receptor to CNS	Carries signals FROM receptors (sensory organs, skin) TO the CNS. Long dendrite from receptor, cell body in dorsal root ganglion, axon to spinal cord.
Interneuron (relay neuron)	CNS (brain and spinal cord)	Connects sensory and motor neurons. Processes and integrates signals within the CNS. Vast network in brain — basis of thought and integration.
Motor neuron	CNS → effector (PNS)	Carries signals FROM CNS TO effectors (muscles, glands). Cell body in spinal cord, long myelinated axon to muscle.

PROCESS: Resting Potential (−70mV) — How it is Maintained

1. Na^+/K^+ ATPase pump (active transport): continuously pumps 3 Na^+ OUT and 2 K^+ IN — using ATP. Net effect: more positive ions outside the membrane.
2. K^+ leak channels: K^+ tends to leak back OUT down its concentration gradient through open K^+ channels — contributing to negative interior.
3. Large negatively charged proteins inside the cell: cannot cross the membrane, contribute to negative interior charge.
4. Net result: inside of membrane is ~−70mV relative to outside. This is the resting potential — the membrane is POLARISED.

PROCESS: Action Potential — Step by Step

1. STIMULUS: A stimulus (above threshold) depolarises a small region of membrane.
2. DEPOLARISATION: Voltage-gated Na^+ channels OPEN → Na^+ rushes IN (down its electrochemical gradient) → inside becomes more positive → membrane potential rises from −70mV to +40mV.
3. REPOLARISATION: Voltage-gated K^+ channels OPEN (slightly delayed) → K^+ rushes OUT → inside becomes more negative → membrane potential falls back toward −70mV.
4. HYPERPOLARISATION: K^+ channels close slowly — K^+ overshoots, making inside briefly more negative than −70mV.
5. REFRACTORY PERIOD: Na^+ channels are inactivated and cannot reopen immediately — no new action potential can be generated. This ensures UNIDIRECTIONAL propagation of the signal along the axon.
6. RESTORATION: Na^+/K^+ pump restores original ion concentrations (takes time — not immediately necessary for each AP).

KEY DEFINITIONS

All-or-nothing principle: an action potential either fires at full amplitude or not at all. Stimulus must reach THRESHOLD (typically around −55mV). If threshold is not reached → no action potential.

Myelinated axons and saltatory conduction: Schwann cells wrap around the axon, forming the myelin sheath (insulating lipid layer). Gaps in myelin = nodes of Ranvier. The action potential 'jumps' from node to node (saltatory conduction) → MUCH FASTER than in unmyelinated axons. Allows high-speed signals over long distances.

Synaptic transmission sequence: Action potential arrives at presynaptic terminal → Ca^{2+} channels open → Ca^{2+} enters → synaptic vesicles fuse with membrane (exocytosis) → neurotransmitter released into synaptic cleft → neurotransmitter binds to receptors on postsynaptic membrane → ion channels open → EPSP (excitatory, depolarises) or IPSP (inhibitory, hyperpolarises) generated → neurotransmitter removed by enzymatic breakdown (e.g. acetylcholinesterase) or reuptake.

Summation: temporal summation (rapid, repeated stimulation from one neuron) and spatial summation (simultaneous stimulation from multiple neurons) — both can bring postsynaptic membrane to threshold.

EXAM TIP

Resting potential = −70mV. This is the starting state. Action potential peaks at +40mV. Know the VALUES.

The refractory period serves TWO important functions: (1) ensures UNIDIRECTIONAL propagation (signal cannot go backward — that region is in refractory period), (2) limits the frequency of action potentials (determines max firing rate).

Multiple sclerosis: autoimmune disease where myelin sheath is damaged → saltatory conduction disrupted → slower/failed signal transmission → muscle weakness, sensory problems.

C3.1 — Integration of Body Systems

CNS/PNS · Reflex arc · Brain · Circadian rhythm · Endocrine system · Feedback

KEY DEFINITIONS

Central Nervous System (CNS): brain + spinal cord. Receives, processes, and integrates information. Protected by bone (skull + vertebrae) and meninges.

Peripheral Nervous System (PNS): all nerves outside the CNS. Somatic division: voluntary — controls skeletal muscle. Autonomic division: involuntary — controls heart, smooth muscle, glands. Autonomic has sympathetic (fight-or-flight) and parasympathetic (rest-and-digest) branches that oppose each other.

Brain region	Location	Key functions
Cerebrum (cerebral cortex)	Largest — fills most of the skull — two hemispheres	Conscious thought, voluntary movement, sensory perception, language, memory, personality, learning
Cerebellum	At the back of the brain, below the cerebrum	Coordination of movement, balance, posture, fine motor control — receives input from muscles and visual system
Medulla oblongata (brainstem)	Lowest part of brain — connects to spinal cord	Controls autonomic functions: breathing rate, heart rate, blood pressure, swallowing, vomiting
Hypothalamus	Deep within brain — below thalamus	Master regulator of homeostasis: body temperature, blood osmolarity, hunger/satiety, sleep-wake cycle. Links nervous and endocrine systems — controls pituitary gland.

PROCESS: Pain Withdrawal Reflex Arc — Sequence

1. RECEPTOR: Sensory receptor in skin detects the painful stimulus (e.g. pin prick) — generates a signal.
2. SENSORY NEURON: Signal travels along sensory neuron to the spinal cord (via dorsal root).
3. INTERNEURON: Signal processed in spinal cord — interneuron passes signal to motor neuron (AND simultaneously sends signal up to brain for conscious awareness — but this is slower and not required for the reflex).
4. MOTOR NEURON: Signal travels along motor neuron (via ventral root) to the effector.
5. EFFECTOR: Muscle contracts — limb withdraws from the stimulus.
6. KEY POINT: The reflex arc is integrated in the SPINAL CORD — response is involuntary and FASTER than a voluntary response because the brain is bypassed (brain receives the signal but the withdrawal has already occurred).

KEY DEFINITIONS

Circadian rhythm: ~24-hour biological clock regulating sleep-wake cycles, hormone secretion, body temperature, and many metabolic processes. Coordinated by the suprachiasmatic nucleus (SCN) in the hypothalamus, which receives light input from retinal ganglion cells. Melatonin (produced by pineal gland in darkness) promotes sleep. Disrupted by shift work, jet lag, artificial light.

Endocrine system: glands that secrete HORMONES directly into the bloodstream. Hormones act on target cells that have specific receptors for that hormone. Slower than nervous system but longer-lasting effects.

Negative feedback: a regulatory mechanism where the OUTPUT of a process OPPOSES the change that triggered it — returning the system to the set point. Used for homeostasis (blood glucose, thermoregulation, blood pressure).

Positive feedback: the OUTPUT AMPLIFIES the initial change. Less common in biology — used when a rapid response to completion is needed (e.g. oxytocin during childbirth: contractions → more oxytocin → stronger contractions → delivery).

EXAM TIP

Reflex arc vs voluntary action: REFLEX is involuntary, processed in SPINAL CORD, FASTER (fewer synapses). Voluntary action is conscious, processed in BRAIN, slower.

Negative feedback is the basis of ALL homeostatic mechanisms. Always describe: (1) the set point, (2) the deviation detected, (3) the corrective response, (4) how it returns to set point.

C3.2 — Defence Against Disease

Non-specific · Innate immunity · Adaptive immunity · Antibodies · Vaccines · Antibiotics · Zoonoses

The immune system protects against pathogens — organisms or agents that cause disease. It operates through non-specific (innate) defences (rapid, general) and specific (adaptive) immunity (slower, targeted, with memory).

Defence type	Timing	Specificity	Memory	Key components
Non-specific (physical/chemical barriers)	Immediate — always present	Non-specific — effective against all pathogens equally	No memory	Skin (physical barrier), mucus (traps pathogens), cilia (sweep mucus away), stomach acid (pH 2 — kills most bacteria), lysozyme in tears/saliva
Innate immunity (cellular)	Hours — fast	Non-specific	No memory	Neutrophils (phagocytes — destroy bacteria rapidly), macrophages (phagocytes + antigen-presenting cells), natural killer cells (destroy virus-infected cells), inflammation response, complement system, fever

Adaptive immunity (humoral)	Days — slower	SPECIFIC to each antigen — highly targeted	YES — immunological memory (memory B cells)	B lymphocytes → plasma cells → antibodies (immunoglobulins). Each B cell recognises ONE specific antigen.
Adaptive immunity (cell-mediated)	Days — slower	SPECIFIC to each antigen	YES — memory T cells	T helper cells (Th) — activate B cells and cytotoxic T cells, release cytokines. T cytotoxic cells (Tc) — kill virus-infected cells and tumour cells directly. T regulatory cells — suppress immune response when infection cleared.

PROCESS: Adaptive Immune Response — Step by Step

1. PATHOGEN ENTRY: Pathogen enters the body and displays antigens on its surface (or on infected cell's MHC molecules).
2. ANTIGEN PRESENTATION: Macrophages engulf pathogen → present antigen fragments on their surface (MHC II) → become antigen-presenting cells (APCs).
3. T HELPER CELL ACTIVATION: T helper cell with complementary receptor binds to APC → activated → releases cytokines → clonal expansion of specific T helper cells → stimulate B cells and cytotoxic T cells.
4. B CELL ACTIVATION (clonal selection): Specific B cell whose surface immunoglobulin matches the antigen binds it (AND receives T helper cytokine signal) → activated → CLONAL EXPANSION (rapid division of activated B cell).
5. PLASMA CELLS: Most clonal B cells differentiate into plasma cells → secrete large quantities of ANTIBODIES (same specificity as original B cell receptor) into blood.
6. MEMORY CELLS: Some B cells and T cells become MEMORY CELLS — long-lived, ready to respond rapidly if the same antigen is encountered again → basis of immunological memory and vaccine protection.
7. ANTIBODY ACTION: Antibodies bind to antigens → neutralisation (block pathogen from infecting cells), agglutination (clump pathogens for easier phagocytosis), complement activation → pathogen destruction.

Topic	Key facts for exam
Antibody structure	Y-shaped glycoprotein. 4 polypeptide chains: 2 heavy + 2 light. Variable regions (tips of Y) — unique to each antibody, bind specific antigen. Constant region (tail) — same in all antibodies of same class; determines effector function. 2 identical antigen-binding sites.
Primary vs secondary immune response	Primary: first exposure to antigen. Slow (days), low antibody titre, mostly IgM produced. Memory cells formed. Secondary: second exposure to same antigen. FASTER (hours), HIGHER antibody titre, mostly IgG, longer-lasting. Due to memory cells — larger population of specific cells ready to respond.
HIV/AIDS	HIV (Human Immunodeficiency Virus) infects and DESTROYS T HELPER (CD4+) cells → immune system progressively collapses. AIDS: CD4+ count drops below 200 cells/μL → unable to fight opportunistic infections (Pneumocystis pneumonia, Kaposi's sarcoma, CMV). Transmitted via blood, sexual contact, breast milk.
Antibiotics	Target bacterial-specific structures: cell wall synthesis (penicillin, ampicillin — target peptidoglycan), protein synthesis at 70S ribosomes (streptomycin, erythromycin). NOT effective against viruses (no cell wall, use host ribosomes). Antibiotic resistance: natural selection of resistant strains (misuse/overuse of antibiotics). MRSA — methicillin-resistant S. aureus.
Vaccines	Introduce ANTIGENS (weakened/killed pathogen, subunit, mRNA) to stimulate IMMUNE RESPONSE and create MEMORY CELLS — without causing disease. When real pathogen encountered: rapid secondary immune response prevents disease. Types: live attenuated (MMR), inactivated (flu), subunit (hepatitis B), mRNA (COVID-19).
Zoonoses	Infectious diseases transmitted from animals to humans. Examples: COVID-19 (bat → humans?), influenza (birds/pigs → humans), Ebola (bats → humans), malaria (Plasmodium parasite via Anopheles mosquito), Lyme disease (deer tick → humans). Risk factors: deforestation, wet markets, intensive farming.

EXAM TIP

Primary vs secondary response: SECONDARY is FASTER and produces MORE antibodies — because of memory cells. Know the graph shape: primary = slow rise, plateau, slow decline. Secondary = rapid steep rise, higher peak, slower decline.

Antibiotics vs antivirals: antibiotics target bacteria-specific structures (cell wall = peptidoglycan, 70S ribosomes). Viruses have neither — they use the HOST cell's machinery. That is why antibiotics do not work on viral infections.

C4.1 — Populations & Communities

Sampling · Population growth · Interspecific relationships · Invasive species · Chi-squared · Succession

KEY DEFINITIONS

Population: all individuals of ONE species in a defined area at a given time.

Community: ALL populations of different species living in the same area and interacting with each other.

Ecosystem: community + its abiotic (non-living) environment.

Sampling method	How it works	Best used for	Limitation
Random quadrats	Quadrats (frames of known area) placed at random coordinates in the study area. Count/estimate organisms inside.	Plants, sessile animals, slow-moving organisms	Random placement must be TRULY random (use random number generator). Cannot be used for mobile animals.
Belt transect	Quadrats placed at regular intervals along a line across the habitat.	Studying changes across an environmental gradient (e.g. from sea to land, forest edge to interior)	Time-consuming. Only samples along the transect line.
Mark-recapture (Lincoln-Petersen index)	Capture sample, mark, release. Wait. Capture again. Count marked in second sample. $N = (n_1 \times n_2) / m$ where N=population, n_1=first capture, n_2=second capture, m=marked in second sample.	Mobile animals that can be marked without harm	Assumes: marks don't affect survival, no births/deaths/migration between captures, marks don't rub off, equal probability of capture.

Growth type	Shape	Conditions	Key features
Exponential (J-curve)	J-shaped — rapid acceleration	Unlimited resources, no predators, no disease — rare in nature	Growth rate proportional to population size. Biotic potential (rmax). Cannot continue indefinitely.
Logistic (S-curve)	S-shaped — slows and levels off at K	Limited resources — as population grows, resources become scarcer	Growth rate slows as population approaches CARRYING CAPACITY (K) — maximum population the environment can sustainably support. K set by limiting factors.

Interspecific relationship	Effect on species A	Effect on species B	Example
Predation	Positive (food)	Negative (consumed)	Wolf (+) and elk (−)
Competition	Negative (resources competed for)	Negative (resources competed for)	Two plant species competing for light and nutrients
Mutualism	Positive	Positive	Clownfish and sea anemone; mycorrhizal fungi and plant roots; nitrogen-fixing bacteria and legumes
Commensalism	Positive (benefits)	Neutral (unaffected)	Epiphytes growing on trees; barnacles on whales
Parasitism	Positive (gains nutrients/shelter)	Negative (harmed but not immediately killed)	Tapeworm and human; mistletoe and tree; flea and dog

KEY DEFINITIONS

Endemic species: found ONLY in one specific geographic area — nowhere else in the world. May be due to isolated island/mountain/lake habitat. Examples: lemurs in Madagascar, Darwin's finches on Galapagos islands, tuatara in New Zealand.

Invasive species: non-native species introduced to a new area (deliberately or accidentally) that spreads rapidly, harms native species, and alters ecosystem function. They often lack natural predators/competitors in the new environment. Examples: cane toads (Australia), zebra mussels (Great Lakes, USA), grey squirrel (UK).

Chi-squared test (χ^2): statistical test to determine if observed frequencies differ significantly from expected frequencies. In ecology: are species distributed non-randomly? $\chi^2 = \Sigma(O-E)^2/E$. Compare to critical value at specified degrees of freedom and significance level.

Primary succession: colonisation of bare, lifeless substrate (e.g. volcanic rock, sand dunes) — starting from zero. Pioneer species (e.g. lichens, mosses) begin soil formation. Slow process. Eventually reaches climax community.

Secondary succession: recovery of a disturbed area where soil already exists (e.g. after fire, after farming cessation). Faster than primary — soil and seed bank already present.

EXAM TIP

Lincoln-Petersen mark-recapture: $N = (n_1 \times n_2) / m$. Know this formula. Also know all 4 ASSUMPTIONS — these are frequently examined. Missing even one assumption loses a mark.

Density-dependent limiting factors: affected by population density (competition, predation, disease, parasitism — all worsen as population increases). Density-independent: unaffected by density (natural disasters, temperature extremes, floods).

C4.2 — Transfers of Energy & Matter

Trophic levels · Energy efficiency · Productivity · Carbon cycle · Keeling Curve

KEY DEFINITIONS

Carbon fixation: the process of converting inorganic CO_2 into organic carbon compounds. Primarily by photosynthesis (RuBisCO enzyme, Calvin cycle), also by chemoautotrophs.

Gross Primary Productivity (GPP): total rate of photosynthesis in an ecosystem — all organic matter produced by producers per unit time per unit area.

Net Primary Productivity (NPP): GPP minus the respiratory losses of the producers. $NPP = GPP - R_p$ (plant respiration). NPP = organic matter available to primary consumers (herbivores).

Energy transfer efficiency: ~10% of energy at one trophic level is transferred to the next. ~90% is lost as heat (from respiration), in undigested material (faeces), and in non-consumed parts.

Keeling Curve: the continuous record of atmospheric CO_2 concentration measured at Mauna Loa Observatory, Hawaii, since 1958 by Charles Keeling. Shows ANNUAL OSCILLATION (seasonal photosynthesis in Northern Hemisphere) superimposed on a LONG-TERM RISING TREND (due to fossil fuel combustion and deforestation). Definitive evidence for human impact on atmospheric CO_2.

Carbon cycle process	Net effect on atmospheric CO_2	Details
Photosynthesis	Removes CO_2 from atmosphere	CO_2 fixed into organic compounds by plants, algae, cyanobacteria. Major annual sink.
Respiration (all organisms)	Adds CO_2 to atmosphere	Aerobic respiration in all living organisms releases CO_2 from organic compounds.
Decomposition	Adds CO_2 to atmosphere	Saprotrophic microorganisms break down dead organic matter → release CO_2 (and methane in anaerobic conditions).
Combustion of fossil fuels	Adds CO_2 (ancient carbon) to atmosphere	Burns coal, oil, natural gas → releases CO_2 fixed millions of years ago. Major contributor to rising CO_2.
Ocean uptake	Removes CO_2 from atmosphere	CO_2 dissolves in seawater → carbonic acid (→ acidification) → some fixed by marine phytoplankton or precipitated as calcium carbonate.
Deforestation	Net release of CO_2	Trees burned or decomposed → CO_2 released. AND loss of trees reduces photosynthesis (future CO_2 fixation).

KEY DEFINITIONS

Nitrogen cycle: nitrogen is essential for amino acids and nucleic acids. N_2 gas cannot be used directly by most organisms. Key processes: Nitrogen fixation ($N_2 \rightarrow NH_3/NH_4^+$ — by Rhizobium bacteria in legume root nodules, or lightning, or Haber process). Nitrification ($NH_4^+ \rightarrow NO_2^- \rightarrow NO_3^-$ by nitrifying bacteria — Nitrosomonas, Nitrobacter). Assimilation (NO_3^- absorbed by plant roots → incorporated into amino acids). Denitrification ($NO_3^- \rightarrow N_2$ by denitrifying bacteria in anaerobic conditions — returns N_2 to atmosphere).

EXAM TIP

Energy transfer: '10% efficiency' = 90% LOST. Lost as: heat (from respiration), uneaten material, undigested material (faeces). This explains why food chains rarely exceed 4–5 trophic levels (not enough energy remains).

Biomass pyramid vs energy pyramid: Energy pyramids are ALWAYS pyramid-shaped (energy always lost). Biomass pyramids can sometimes be inverted (e.g. ocean: phytoplankton biomass < zooplankton — phytoplankton reproduce so fast their standing stock is low but they have high productivity).

🔗 LINKS TO OTHER TOPICS

C1.1 Enzymes → C1.2 (enzymes catalyse every respiration step), C1.3 (RuBisCO in Calvin cycle), C3.2 (enzymes involved in complement and antibody production)

C1.2 Respiration → B2.2 (mitochondria), A1.2 (NADH — uses NAD^+ made from nucleotides)

C1.3 Photosynthesis → B3.1 (CO_2 enters via stomata), B2.2 (chloroplasts), C4.2 (primary productivity, carbon cycle)

C2.2 Neural signalling → C3.1 (reflex arc, brain function), D3.3 (hypothalamus in homeostasis)

C3.2 Immunity → A2.2 (lymphocytes are cells), B1.2 (antibodies are proteins), A3.1 (viruses are not cells)

C4.1/C4.2 Ecology → D4.2 (ecosystem stability), D4.3 (climate change disrupts ecosystems and carbon cycle)

THEME C — EXAM READINESS CHECKLIST

C1.1: **ENZYMES & METABOLISM**

- [x] Induced fit model
- [x] How enzymes lower activation energy
- [x] 4 factors affecting rate — explain mechanism for each
- [x] Competitive vs non-competitive inhibition — 3 differences

C1.2: **CELL RESPIRATION**

- [x] ATP structure and ATP cycle
- [x] 4 stages of aerobic respiration — location, inputs, outputs
- [x] How ETC makes ATP (chemiosmosis)
- [x] Why O_2 is needed
- [x] Anaerobic pathways and WHY they occur
- [x] Compare aerobic vs anaerobic ATP yield

C1.3: **PHOTOSYNTHESIS**

- [x] Light-dependent reactions — location, inputs, outputs, events
- [x] Calvin cycle — location, 3 stages (fixation, reduction, regeneration)
- [x] Chromatography — Rf value
- [x] Action vs absorption spectrum
- [x] 3 limiting factors and their mechanisms

C2.2: **NEURAL SIGNALLING**

- [x] Neuron structure
- [x] Resting potential — how maintained (Na^+/K^+ pump)
- [x] Action potential — 5 steps
- [x] All-or-nothing
- [x] Saltatory conduction
- [x] Synapse sequence — 7 steps
- [x] EPSP vs IPSP

C3.1: **INTEGRATION OF BODY SYSTEMS**

- [x] CNS vs PNS
- [x] Brain regions and functions
- [x] Reflex arc — 5 components in order
- [x] Circadian rhythm — what regulates it
- [x] Negative vs positive feedback

C3.2: **DEFENCE AGAINST DISEASE**

- [x] Non-specific vs specific immunity
- [x] Adaptive immune response — 7 steps
- [x] Antibody structure
- [x] Primary vs secondary immune response — differences
- [x] HIV mechanism
- [x] Antibiotics — why not effective on viruses
- [x] Vaccines — mechanism

C4.1: **POPULATIONS & COMMUNITIES**

☑ 3 sampling methods — assumptions of mark-recapture
☑ Lincoln-Petersen formula
☑ J-curve vs S-curve — explain difference
☑ 5 interspecific relationships
☑ Succession — primary vs secondary

C4.2: **TRANSFERS OF ENERGY & MATTER**

☑ GPP vs NPP
☑ ~10% energy transfer efficiency
☑ Carbon cycle — 6 processes
☑ Keeling Curve — what it shows and its significance
☑ Nitrogen cycle — 4 key processes